

BYOP: AI Language Detection System

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Project Category: Natural Language Processing (NLP)

Project Overview

The **AI Language Detection System** is a high-performance Command Line Interface (CLI) application developed to identify the linguistic origin of digital text. By implementing advanced statistical probability models, the system evaluates input—ranging from single phrases to comprehensive text files—to determine the most probable language, providing a transparent breakdown of its analytical confidence.

Core Features

- **Real-time Analysis:** Instantaneous detection of manually entered text strings.
- **File System Integration:** Support for batch processing and auditing of .txt documents.
- **Probability Distribution:** Displays the primary language match alongside three alternative candidates with precise percentage-based confidence scores.
- **Validation Logic:** Automated warnings for insufficient data lengths to ensure result integrity.
- **Data Persistence:** Integrated history logging that tracks the last 10 operations via a *history.json* architecture.

Getting Started

Technical Requirements

This system requires a stable installation of **Python 3.x**.

Installation Procedure

1. Download or clone the project repository to your local machine.
2. Install the necessary dependency using the following command:

```
pip install langdetect
```

Application Execution

To initialize the system, navigate to the project directory and execute:

```
python main.py
```

Operation Instructions

Upon initialization, the system presents a formal management menu:

- **Analyze text input:** Used for direct manual entry and testing.
- **Analyze a .txt file:** Used for processing external file paths.
- **View analysis history:** Accesses the local data log for previous sessions.
- **Exit:** Terminates the application and saves the current state.

Methodology Note

The engine utilizes **N-gram frequency analysis** coupled with a **Naive Bayes Classifier**. It references pre-trained language profiles to match linguistic patterns. For consistency in academic testing, a deterministic seed is utilized within the detection factory to ensure reproducible results.